MRP: Mini Research Problem

Sentiment Analysis Model Comparison

Name: Mingjie Qiu

ID: 313481

Professor: Dr. Aleksandr Petiushko

Course: MSCS2201-2: Artificial Intelligence

Term: Winter 2026

Overview

- Motivation
  - Sentiment analysis is a fundamental task in **Natural Language Processing (NLP)** used to automatically determine whether text expresses **positive or negative opinions**. It has many real-world applications such as:
    - analyzing customer feedback
    - monitoring social media sentiment
    - understanding public opinion on products and services
  - However, different neural network architectures handle language differently, and their performance can vary significantly.
- Main Idea
  - This project compares **three deep learning architectures** for sentiment classification on movie reviews:
    - **GRU (Gated Recurrent Unit)** – an efficient recurrent neural network
    - **BiLSTM (Bidirectional Long Short-Term Memory)** – captures context from both directions
    - **DistilBERT** – a transformer-based pretrained language model
  - All models are trained and evaluated on the **IMDb movie review dataset**
- Final Result:
  - Transformer models capture contextual meaning better than traditional RNN architectures.
  - DistilBERT achieves the highest accuracy (85.2%).

LITERATURE REVIEW

What is Sentiment Analysis?

- Sentiment analysis is a Natural Language Processing (NLP) task that identifies emotional tone in text.
- It determines whether text expresses positive, negative, or neutral sentiment.
- Common applications include product reviews, social media monitoring, and customer feedback analysis.
- Companies use sentiment analysis to automatically understand public opinion.
- Machine learning models learn patterns in language that correlate with sentiment.

Recurrent Neural Networks

Early deep learning approaches for Natural Language Processing relied heavily on **Recurrent Neural Networks (RNNs)** because they are designed to process **sequential data such as sentences and documents**. Two widely used RNN variants include:

- GRU (Gated Recurrent Unit)
  - Introduced to address the **vanishing gradient problem** in traditional RNNs.
  - Uses gating mechanisms to control information flow through the network.
  - More computationally efficient compared to LSTM.
- BiLSTM (Bidirectional Long Short-Term Memory)
  - Processes sequences in **both forward and backward directions**.
  - Captures contextual information from **both past and future words** in a sentence.
  - Commonly used in sentiment analysis and other text classification tasks.

However, RNN-based models often struggle with **long-range dependencies and slow training due to sequential processing**.

Transformer-Based Models

Recent advances in NLP have been driven by the **Transformer architecture.** Transformers replace recurrence with **self-attention mechanisms**, allowing models to process entire sentences simultaneously while learning relationships between words.

Key advantages include:

- Ability to capture **long-distance word relationships**
- **Parallel computation**, leading to faster training
- Strong performance across many NLP tasks

One widely used transformer model is:

- DistilBERT is a **compressed version of BERT** that maintains high accuracy while reducing computational cost.

Because of these benefits, transformer-based models have become the **state-of-the-art approach for sentiment analysis and other NLP tasks**.

DATASET

IMDb Movie Review Dataset

- Introduction
  - The IMDb dataset is a widely used benchmark for sentiment classification tasks.
  - It contains thousands of movie reviews written by real users.
  - Each review is labeled as either positive or negative sentiment.
  - Reviews often contain complex sentences, sarcasm, and emotional language.
  - These characteristics make the dataset realistic and challenging for NLP models.
- Feature
  - Total dataset size: 50,000 reviews.
  - Training set: 25,000 labeled reviews.
  - Test set: 25,000 labeled reviews.
  - Binary classification: Positive vs Negative.
  - Reviews vary significantly in length and writing style.
- Link
  - IMDB Dataset

Datasets: stanfordnlp/imdb | like 362 | Follow Stanford NLP 336 | Dataset card | Data Studio | Files and versions xet | Community 8

Split (3)
train · 25k rows

Search this dataset

text string · *lengths* 52 — 13.7k	label class label 2 classes
I rented I AM CURIOUS-YELLOW from my video store because of all the controversy that surrounded it when it was first released in 1967. I also heard that at first it was seized…	0 neg
"I Am Curious: Yellow" is a risible and pretentious steaming pile. It doesn't matter what one's political views are because this film can hardly be taken seriously on any level. As…	0 neg
If only to avoid making this type of film in the future. This film is interesting as an experiment but tells no cogent story. One might feel virtuous for sitting thru i…	0 neg
This film was probably inspired by Godard's Masculin, féminin and I urge you to see that film instead. The film has two strong elements and those are, (1) the realistic…	0 neg
Oh, brother...after hearing about this ridiculous film for umpteen years all I can think of is that old Peggy Lee song.. "Is that all there is??" ...I was just an early tee…	0 neg
I would put this at the top of my list of films in the category of unwatchable trash! There are films that are bad, but the worst kind are the ones that are unwatchable but you are…	0 neg
Whoever wrote the screenplay for this movie obviously never consulted any books about Lucille Ball, especially her autobiography. I've never seen so many mistakes in a biopic,…	0 neg
When I first saw a glimpse of this movie, I quickly noticed the actress who was playing the role of Lucille Ball. Rachel York's portrayal of Lucy is absolutely awful. Lucille Ball was…	0 neg
Who are these "They"- the actors? the filmmakers? Certainly couldn't be the audience- this is among the most air-puffed productions in existence. It's the kind of movie that looks…	0 neg
This is said to be a personal film for Peter Bogdonavitch. He based it on his life but changed things around to fit the characters, who are detectives. These detectives date…	0 neg
It was great to see some of my favorite stars of 30 years ago including John Ritter, Ben	

‹ Previous 1 2 3 ... 250 Next ›

```
-- The SQL console is powered by DuckDB WASM and runs entirel
-- Get started by typing a query or selecting a view from the
SELECT * FROM train LIMIT 10;
```

Subsets and Splits

train 25k | test 25k | unsupervised 50k

Run Query ⌘↵

Saved Queries | Top Community Queries

Positive Keywords Search
Retrieves sample text entries containing positive keywords, providing basic…

Long Reviews Analysis - Top 15 Longest Texts
Retrieves the longest text entries from the training set, which helps identify…

Positive IMDB Reviews from Test Set
Provides basic filtered examples of text data labeled as category 1, useful for…

Dataset Sampling for This Project

- To reduce training time, a subset of the dataset was used.
- Training subset: 5,000 reviews randomly selected.
- Testing subset: 1,000 reviews randomly selected.
- Random sampling ensures balanced class distribution.
- This allows faster experimentation while preserving meaningful evaluation.

APPROACH / METHODOLOGY

Overall Approach

This project evaluates multiple neural network architectures for **sentiment classification** using the IMDb Movie Review Dataset. The workflow follows these steps:

1. Dataset Loading

Reviews were loaded using the Hugging Face Datasets library.

2. Text Preprocessing

- All text is converted to lowercase for consistency.
- Words are tokenized by splitting sentences into tokens.
- A vocabulary dictionary is built using the most frequent words.
- Each word is mapped to an integer index.
- Sequences are padded or truncated to fixed length (128 tokens).

3. Model Training

Three architectures were implemented and trained:

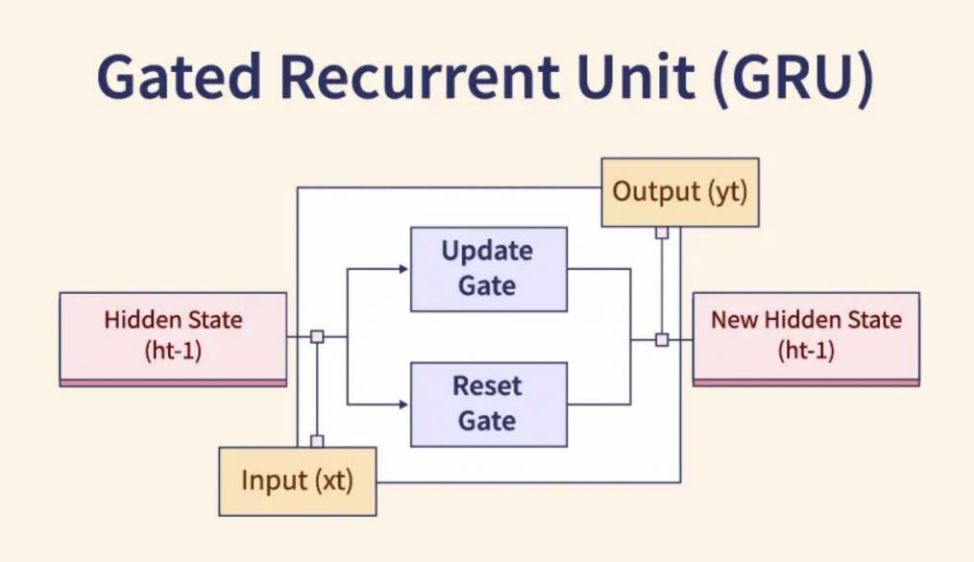
- GRU (Gated Recurrent Unit)
- BiLSTM (Bidirectional LSTM)
- DistilBERT Transformer

4. Evaluation

- Models evaluated using **classification accuracy**
- Additional metrics recorded:
  - training time
  - model parameters
  - efficiency (accuracy per training second)

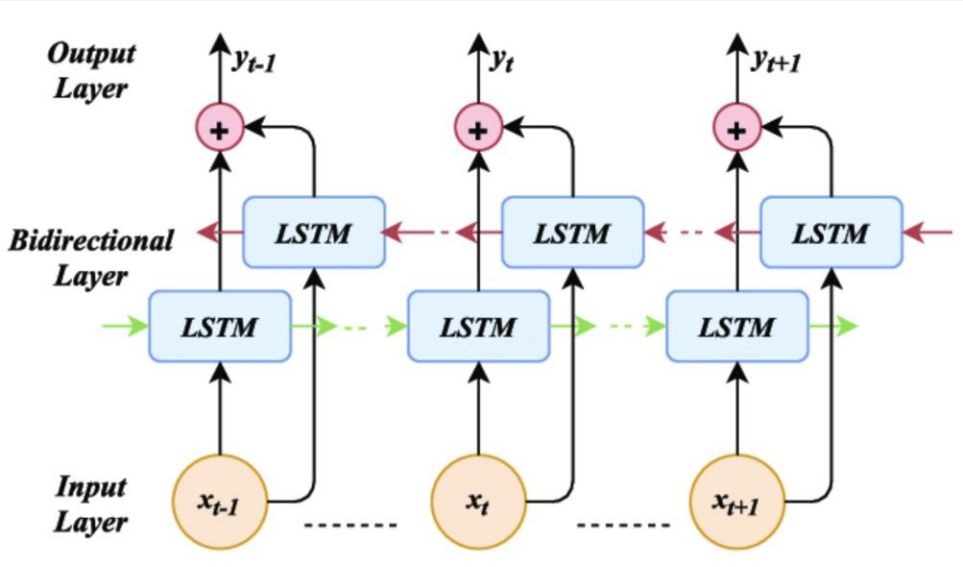
Model 1: GRU (Gated Recurrent Unit)

- Architecture:
  - Bidirectional GRU with 2 layers
  - Embedding dimension: 300
  - Hidden dimension: 600
  - Fully connected layers: 256 → 2 (binary classification)
  - Dropout: 0.5, ReLU activation
- Training Details:
  - Optimizer: AdamW with weight decay 0.01
  - Learning rate scheduler: CosineAnnealingLR
  - Batch size: 32, Epochs: 10
- Key Idea: Captures sequential dependencies and contextual information efficiently, faster than BiLSTM.



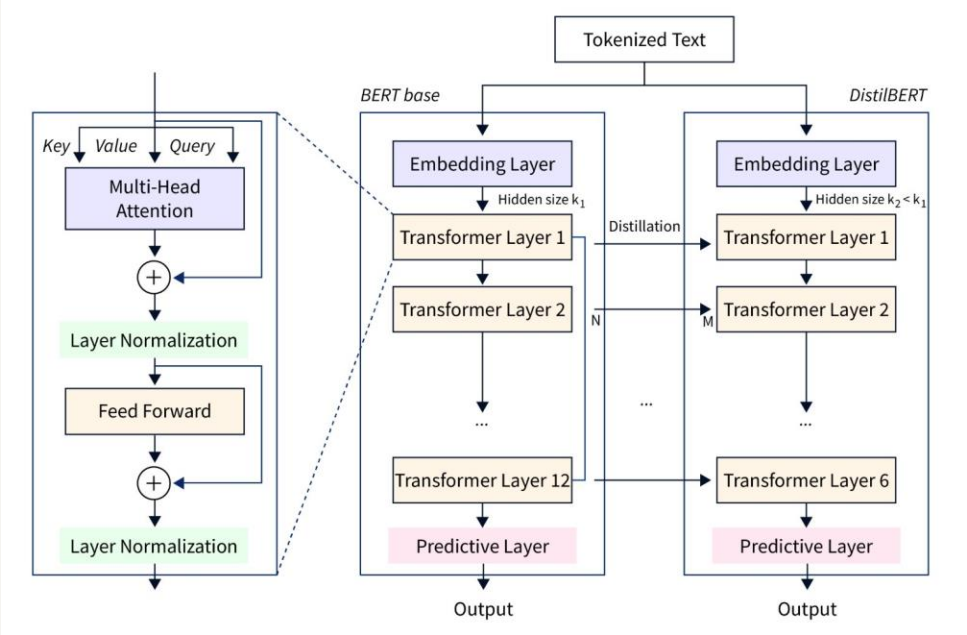
Model 2: BiLSTM (Bidirectional Long Short-Term Memory)

- Architecture:
  - Bidirectional LSTM with 2 layers
  - Embedding dimension: 300
  - Hidden dimension: 600
  - Fully connected layers: 256 → 2
  - Dropout: 0.5, ReLU activation
- Training Details:
  - Optimizer: AdamW with weight decay 0.01
  - Learning rate scheduler: CosineAnnealingLR
  - Batch size: 32, Epochs: 10
- Key Idea: Captures longer dependencies in sequences, but more parameters and slower training than GRU.



Model 3: DistilBERT

- **Architecture:**
  - Pretrained DistilBERT (6-layer transformer)
  - Hidden size: 768
  - Output layer: 2 classes
  - Tokenizer: distilbert-base-uncased
- **Training Details:**
  - Trainer API from Hugging Face
  - Batch size: 8, Epochs: 10
  - Evaluation strategy: per epoch
- **Key Idea:** Pretrained transformer captures deep contextual understanding, highest accuracy despite longer training and model size.



Code Repository

- https://github.com/MingjieQiu/MRP

Video Demonstration

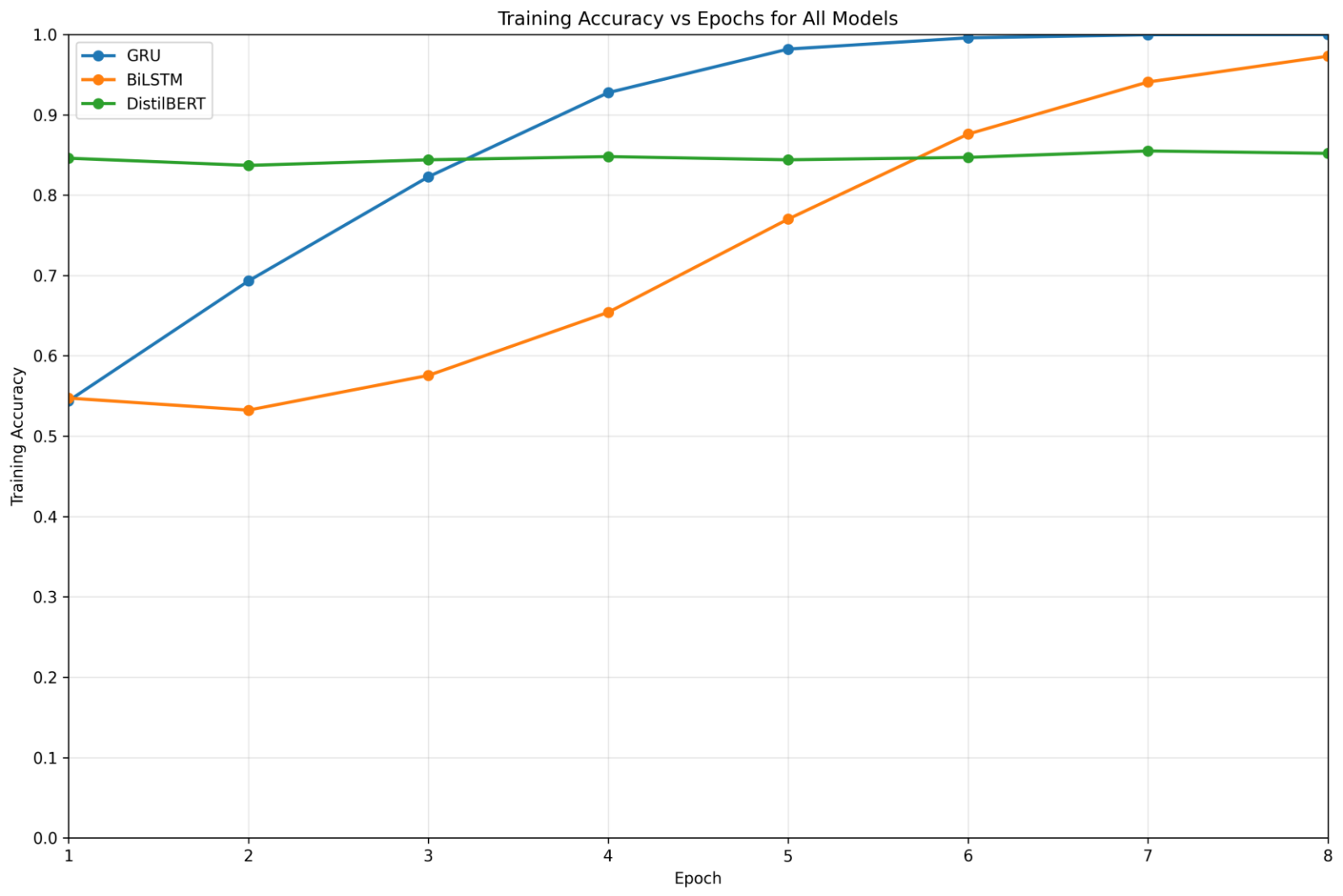
- Mini ResearchProblem - Sentiment Analysis Model Comparison - GRU/BiLSTM/DistilBERT

RESULTS

Training accuracy / epoch

Epoch	GRU	BiLSTM	DistilBERT
1	0.5440	0.5474	0.8460
2	0.6932	0.5324	0.8370
3	0.8228	0.5756	0.8440
4	0.9276	0.6542	0.8480
5	0.9818	0.7702	0.8440
6	0.9958	0.8760	0.8470
7	0.9996	0.9408	0.8550
8	0.9998	0.9730	0.8520

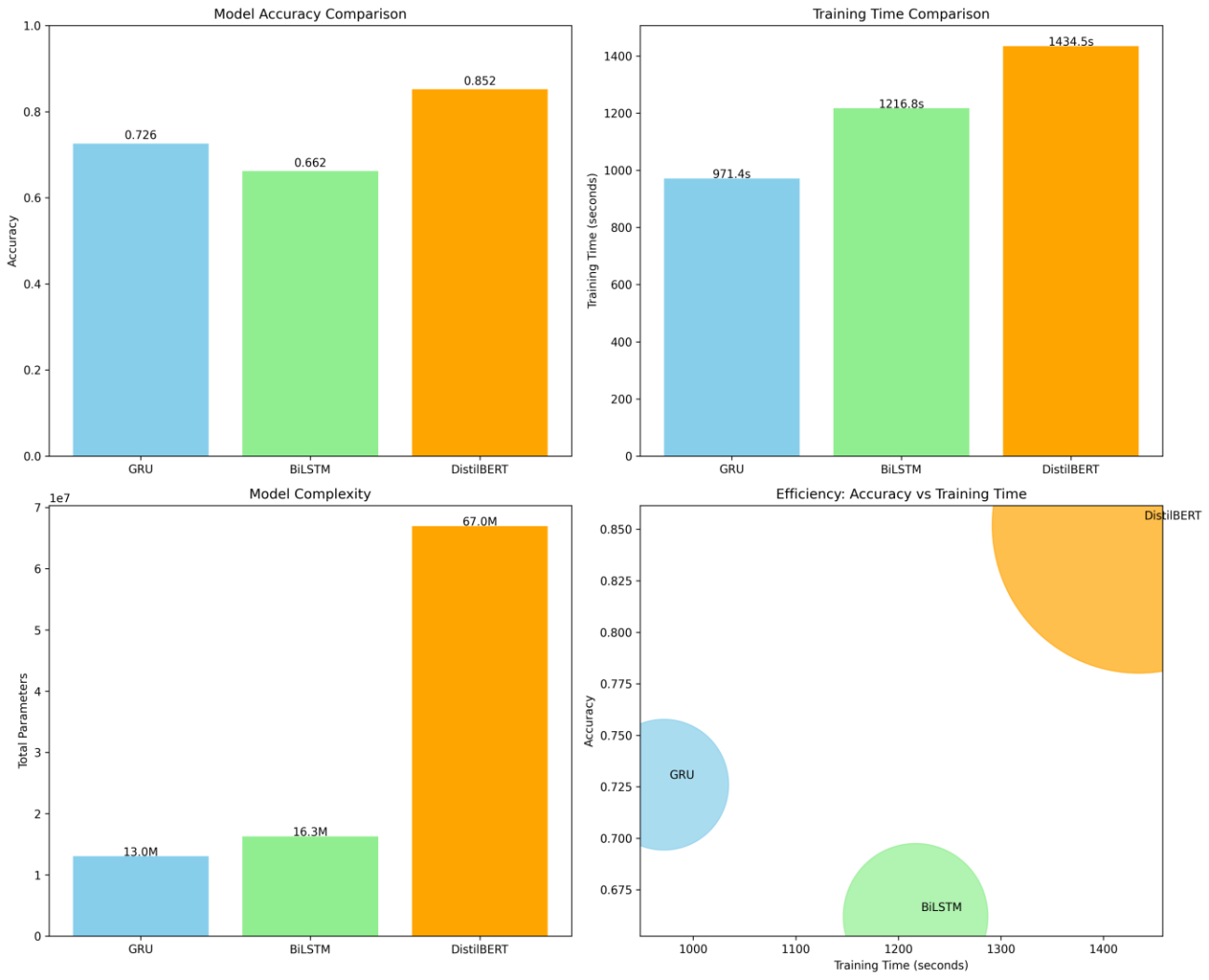
Training Accuracy Visualization



Testing Results

Model	Accuracy	Training Time	Parameters
GRU	72.6 %	971.38 sec	13.04 M
BiLSTM	66.2 %	1216.81 sec	16.28 M
DistilBERT	85.2 %	1434.48 sec	67.95 M

Testing Results Visualization



Comparative Analysis

Training Convergence Behavior

- GRU
  - Demonstrated the **fastest convergence**
  - Training accuracy increased rapidly from **54.4% → 99.98% by epoch 8**
  - Indicates strong capacity to memorize training data.
- BiLSTM
  - **Slower and more gradual convergence**
  - Training accuracy improved from **54.7% → 97.3% by epoch 8**
  - Learning process was more stable but less efficient.
- DistilBERT
  - Achieved **high accuracy from the first epoch (84.6%)**
  - Only **minor improvements across epochs**
  - Shows effectiveness of **pretrained language representations.**
- DistilBERT achieved the highest test accuracy
- GRU performed moderately well with significantly fewer parameters
- BiLSTM showed the lowest performance despite larger model size than GRU

Comparative Analysis

Model Efficiency Trade-offs

- **DistilBERT**
  - Best predictive performance
  - Highest computational cost
  - Largest parameter size
- **GRU**
  - Best **efficiency balance**
  - Faster training and smaller model
  - Suitable for limited computational resources
- **BiLSTM**
  - Higher complexity than GRU but **lower accuracy**
  - Indicates inefficient parameter usage for this dataset.

Key Findings & Insights

1. Transformer Models Provide Superior Accuracy

- DistilBERT achieved **85.2% test accuracy**, outperforming both RNN models.
- The **self-attention mechanism** enables better contextual understanding of text.

2. Evidence of Overfitting in RNN Models

- GRU and BiLSTM training accuracies approached **~100%**.
- However, test accuracies remained significantly lower:
  - GRU: **72.6%**
  - BiLSTM: **66.2%**
  - This suggests **overfitting to the training data**.

3. Pretrained Models Improve Generalization

- DistilBERT showed **smaller gap between training and test performance**.
- Pretraining on large corpora provides **better language understanding**.

Key Findings & Insights

4. Model Complexity vs Performance

- DistilBERT has **~67M parameters**, far larger than RNN models.
- However, **higher parameter count alone does not guarantee performance**, as shown by BiLSTM vs GRU.

5. Practical Deployment Considerations

- **High accuracy applications:** DistilBERT is the best choice.
- **Resource-limited environments:** GRU offers the best trade-off between speed and performance.
- **BiLSTM models may require stronger regularization or larger datasets** to compete.

References

- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I., *“Attention Is All You Need,”* NeurIPS, 2017.
- Devlin, J., Chang, M., Lee, K., & Toutanova, K., *“BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,”* NAACL-HLT, 2019.
- Sanh, V., Debut, L., Chaumond, J., & Wolf, T., *“DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,”* arXiv:1910.01108, 2019.
- Cho, K., van Merriënboer, B., Gulcehre, C., et al., *“Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation,”* EMNLP, 2014.
- Hochreiter, S., & Schmidhuber, J., *“Long Short-Term Memory,”* Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.

Thank You